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Design and Implementation of an Adaptive E-learning Platform based on Quality of Experience for Inclusive Online Education in Low-Resource Settings

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1. Introduction

This report documents the requirements gathering phase for the development of our adaptive mobile e-learning platform. The proposed system is designed to provide a learning experience that automatically adjusts how content is delivered based on the user's real-time internet connection. This ensures that students can stay engaged with their studies even when their connection is unstable or limited.

For this week, we are focused on the first major phase of the System Development Life Cycle (SDLC): Requirements Gathering. In this stage, we collect and analyze information from potential users and stakeholders to define exactly what the system needs to do. This process is vital because the needs identified now will guide every future step of the project, from the initial design and coding to final testing and deployment. Our goal is to ensure the data we collect is accurate and truly reflects the needs of students and instructors in low-resource environments.

1.1 Purpose of This Report

The purpose of this report is to present the requirements gathered from potential users and to outline the methodology used to obtain them. The document covers:

• The specific techniques used to reach stakeholders.

• The process followed to ensure data integrity.

• The raw and cleaned data points collected.

• An assessment of potential user challenges and reluctance.

1.2 Project Background

Students in developing regions often face steep uphill battles when trying to access online education. Common barriers include the high cost of mobile data, low-end hardware, and internet connectivity that fluctuates throughout the day. While many e-learning platforms exist, most are built for high-speed environments and fail to accommodate users with limited resources.

This project seeks to bridge that digital divide by building a "smart" application that adapts to the user. Whether it is by optimizing buffering, switching to audio-only modes, or providing low-resolution alternatives, the system ensures that learning is never cut short by a poor signal. This phase was conducted to understand the lived experiences of these users so we can build a tool that actually works for them.

2. Stakeholder Identification

The success of the Adaptive E-learning Platform depends on identifying all parties that influence or are impacted by the system. In the context of low-resource settings like Cameroon, stakeholders are not limited to human users but also include the technical environment that dictates the app's performance.

2.1 Primary Stakeholders (Direct Users)

These individuals interact directly with the mobile application and provide the functional requirements for the learning experience.

- **Students:** The core beneficiaries of the platform. Their primary interest is accessing educational content (video, audio, text) without interruptions caused by buffering or high data costs. Their requirements focus on usability, data efficiency, and offline functionality.

- **Instructors/Teachers:** Responsible for content creation and delivery. They require a reliable way to upload materials and engage with students. Their requirements focus on content management, ease of uploading, and student progress tracking.

2.2 Secondary Stakeholders (Management & Support)

These entities do not use the app for daily learning but ensure the platform remains operational and relevant.

- **University/School Administrators:** They have an interest in the platform's ability to improve student retention and academic performance. They provide requirements regarding analytics, reporting, and institutional branding.

- **Development Team:** Responsible for the design, coding, and maintenance of the adaptive algorithms. Our interest lies in technical feasibility, system architecture, and successful deployment.

2.3 External Technical Stakeholders (Infrastructure)

In an adaptive system, the environment itself acts as a stakeholder because the software must negotiate with it to function.

- **Internet Service Providers (ISPs - MTN, Orange, Camtel):** These are critical external stakeholders who provide the transmission infrastructure (3G, 4G, and Edge). While we

do not interact with them directly, the application must be Network-Aware. This means the app is designed to monitor the real-time Quality of Service (QoS) provided by these ISPs and adapt the media format accordingly to ensure continuous service.

- **The Network Environment:** The fluctuating bandwidth typical of Sub-Saharan Africa is a silent stakeholder. It imposes the constraints that define our Non-Functional Requirements, specifically Reliability, Scalability, and Performance under high-latency conditions.

The following table summarizes the relationship between each identified stakeholder and their specific influence on the platform's development requirements.

2.4 Stakeholder Impact Summary Table

Stakeholder	Key Interest	Influence on Requirements	Priority
Students	Affordable, stable learning	Functional requirements for Lite Mode and offline downloads.	**High**
Teachers	Effective content delivery	Requirements for easy file uploads and student tracking.	**Medium**
ISPs (MTN/Orange)	Bandwidth management	Technical constraints for network-aware algorithms.	**High**
University Admin	High engagement rates	Requirements for data analytics and student performance reports.	**Low**

3. Requirements Gathering Techniques

Various techniques exist for gathering requirements in system development. These include interviews, focus groups, observation, prototyping, document analysis, and questionnaires or surveys. The choice of technique depends on the nature of the project, the availability of stakeholders, and the type of information needed.

For this project, the survey method was selected as the primary requirements gathering technique. A detailed explanation of why this technique was chosen, and how it was applied, is provided in the sections below.

3.1 Primary Technique: Survey (Google Forms)

A structured survey was designed and distributed to potential users of the system using Google Forms. Google Forms was chosen because it is accessible on mobile devices, does not require the respondent to create an account, and allows for easy collection and export of responses.

The survey was used as the primary technique for the following reasons:

- It allows data to be collected from a large number of people at the same time.
- It is suitable for reaching users who may not be available for face-to-face interviews.
- It can be completed quickly on a smartphone, which is the most common device among the target user group.
- It produces structured responses that are easier to organise and present.
- It protects respondent anonymity, which encourages more honest answers.

3.2 Survey Design

The survey consisted of 21 questions covering the following areas:

- User role and background (student, teacher, or worker)
- Current online learning habits and device usage
- Quality and affordability of internet access
- Past experience with e-learning platforms
- Problems and frustrations experienced during online learning
- Preferred content formats
- Preferred fallback behaviour when internet becomes poor
- Desired features in a new e-learning application
- Importance of offline access and automatic progress saving

The questions used a combination of multiple choice, checkbox (multi-select), Likert scale (1 to 5 rating), and open-ended text responses. This combination allowed both structured and descriptive data to be collected from respondents.

4. Requirements Gathering Process

This section describes how the requirements gathering survey was planned and carried out. It covers the preparation of the survey instrument, the distribution of the survey, and the collection of responses.

4.1 Survey Preparation

The survey was prepared by the requirements gathering team. The questions were designed to directly address the goals of the project — specifically, to understand how users experience online learning under poor internet conditions and what they would need from a new application.

Before distribution, the team reviewed the survey questions to ensure they were clear, simple, and relevant to the target audience. Questions were written in plain English to ensure they could be understood by users with varying levels of education.

4.2 Target Respondents

The survey was targeted at students and teachers who currently use or have used e-learning tools. These individuals represent the primary users of the proposed system. Workers who use digital learning tools were also included as a secondary group.

4.3 Survey Distribution

The survey was distributed digitally through sharing the Google Forms link. Respondents were able to access and complete the survey on any internet-enabled device, including smartphones and laptops. The link was shared through appropriate channels to reach the target user group.

4.4 Response Collection

The survey was active during the period of 18 to 19 April 2026. A total of 59 responses were collected during this period. All responses were automatically recorded in a Google Sheets spreadsheet linked to the form. The data was subsequently exported as a CSV file for review and cleaning.

Detail	Information
Survey tool	Google Forms
Survey period	18 April 2026 to 19 April 2026
Total responses collected	59

Detail	Information
Number of questions	21
Question types used	Multiple choice, Checkbox, Likert scale (1–5), Open-ended text
Data export format	CSV (Comma-Separated Values)

5. Requirements Gathered

This section presents the data collected through the survey. The raw data is presented first, followed by the cleaned data. No interpretation or analysis is included in this section. The data is presented as collected from respondents.

5.1 Raw Data

The raw data refers to the survey responses as they were recorded by Google Forms, before any cleaning or organising was done. The table below provides an overview of what the raw data file contained.

Property	Value
Total rows (one per respondent)	59
Total columns (one per question)	21
File format	CSV
Timestamp range	18 April 2026 11:03 PM to 19 April 2026 (multiple entries)
Language of responses	Mostly English; a small number of French responses observed

As we can see from the raw data, responses were collected over a short period. Some questions, particularly the open-ended ones, showed variation in how respondents expressed the same idea. The raw data also contained missing values in optional fields.

What would make an e-learning app better for you?

63 responses

- Better internet connection
- Smooth playback
- Right videos🙂
- Good network
- Features focusing on adaptive learning
- Messes inter et consumption.
- Easy to use and less Data consumption
- A feedback hub
- If there are real time update on course content

5.1.1 Raw Data Issues Identified

Before the data could be used, the following issues were identified in the raw data:

Issue Identified	Description
Inconsistent capitalisation	The same platform name was entered in different ways, e.g. 'Youtube', 'YouTube', and 'youtube'
Vague or informal platform names	Some respondents listed tools like 'Google', 'ChatGPT', or 'WhatsApp' as e-learning platforms
'No' written in platform name field	Some respondents who selected 'Yes' to using a platform did not provide a platform name
Varied language in open-text	Responses used different words for the same problem, e.g.

Issue Identified	Description
fields	'network', 'network issues', 'network problems', 'slow connexion'
French language responses	A small number of respondents wrote in French, e.g. 'Débit très bas', 'Slow connexion'
Missing values (blank fields)	Some optional questions were left blank by certain respondents

5.2 Cleaned Data

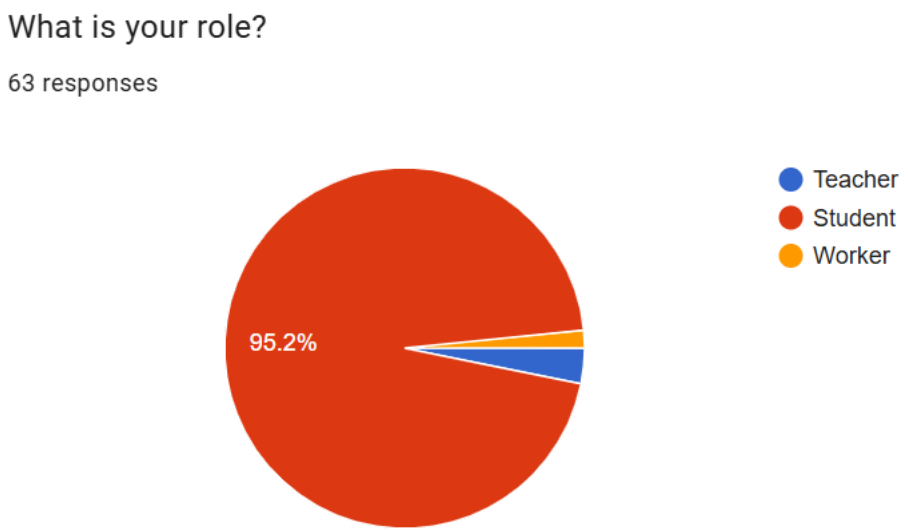
The cleaned data was produced by reviewing the raw responses and organising them into consistent categories. The following steps were taken during the cleaning process:

- Platform names written in different cases were standardised under one label.
- Open-text responses describing the same problem were grouped under a common category label.
- French responses were noted and their meaning recorded in English.
- Blank optional fields were retained as missing values and not substituted with assumptions.
- Informal tools listed as platforms (e.g. WhatsApp, ChatGPT) were retained but noted separately.

5.2.1 Respondent Roles

The following roles were recorded among the 59 respondents:

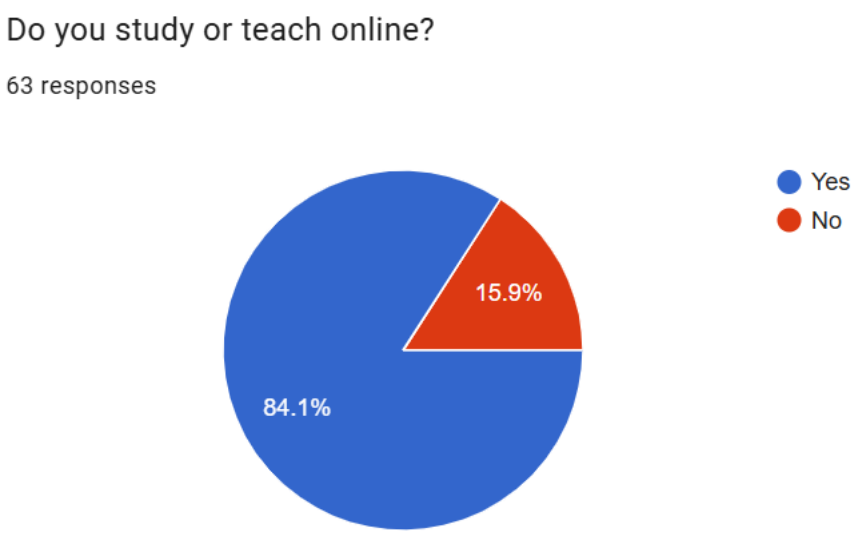
Role	Number of Respondents	Percentage
Student	56	95%
Teacher	2	3%
Worker	1	2%



As we can see, the large majority of respondents are students. This shows that we have to ensure the system is primarily designed for student users, while also accommodating the smaller group of teachers.

5.2.2 Online Learning Participation

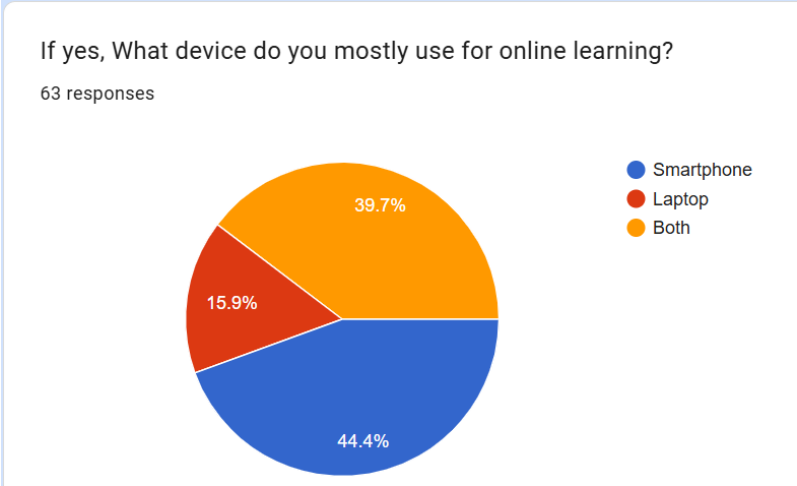
Response	Number of Respondents	Percentage
Yes — currently study or teach online	49	83%
No	10	17%



5.2.3 Devices Used for Online Learning

Device	Number of Respondents	Percentage
Smartphone only	26	44%
Both smartphone and laptop	23	39%
Laptop only	10	17%

As we can see, 83% of respondents use a smartphone either exclusively or alongside a laptop. This shows that we have to design the system to function on mobile devices as a primary requirement.

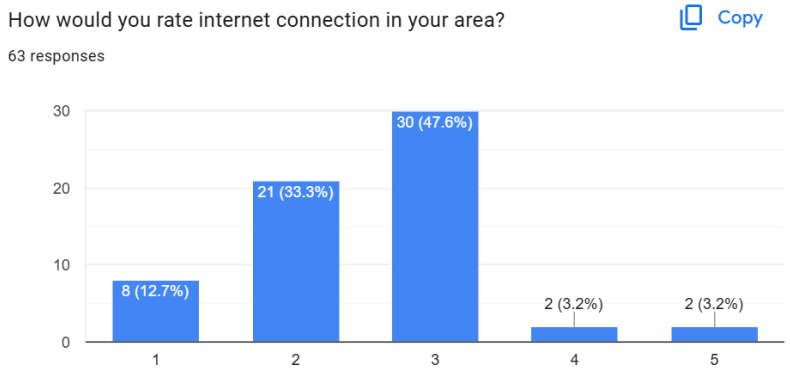


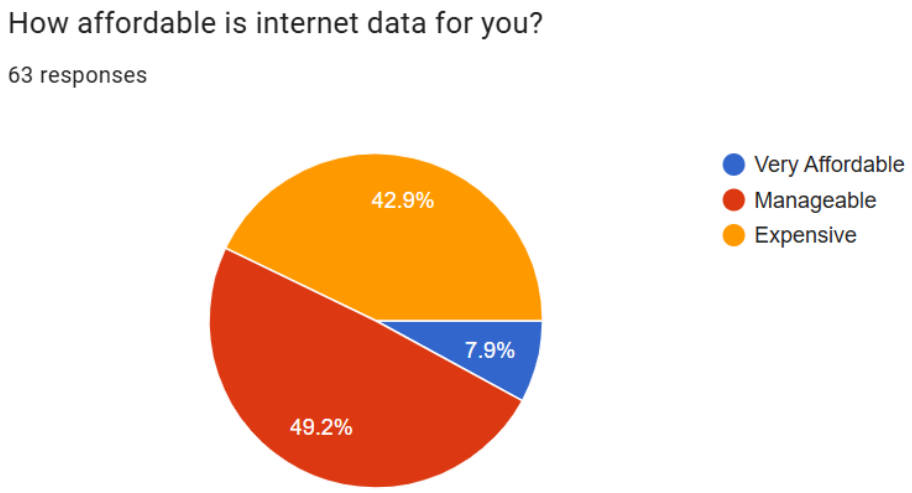
5.2.4 Internet Connection Quality Rating (Scale: 1 = Very Poor, 5 = Excellent)

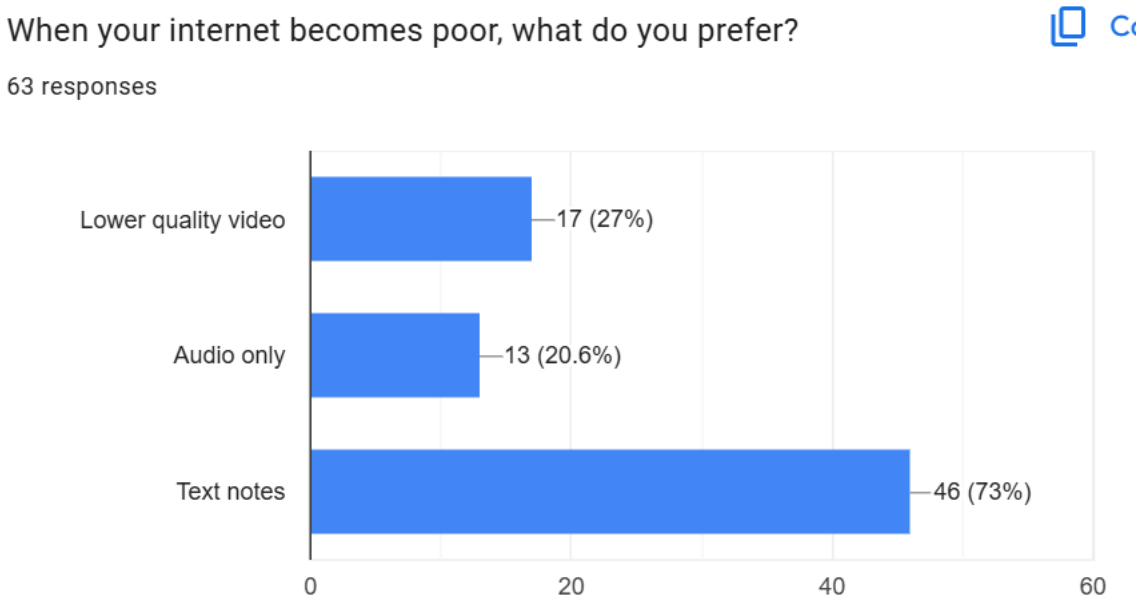
Rating	Meaning	Number of Respondents	Percentage
1	Very poor	8	14%
2	Poor	20	34%
3	Average / Fair	28	47%
4	Good	1	2%

Rating	Meaning	Number of Respondents	Percentage
5	Excellent	2	3%

As we can see, 95% of respondents rated their internet connection as 3 or below. This shows that we have to build a system that treats poor connectivity as the default, not as an exception.







6. User Reluctance Assessment

User reluctance is a primary challenge for any startup, especially when introducing an 'Adaptive' solution in a market with historical infrastructure failures. This report assesses the barriers that cause users to hesitate and explains how our platform converts these risks into trust-building opportunities.

6.1 Financial Risk & Resource Management

Data is a premium resource in our target market. With 44% of respondents classifying data as 'expensive' and 47% as only 'manageable,' users are highly protective of their internet bundles.

- Reluctance Factor: Users worry that an automated system might consume more data than they have budgeted. They are hesitant to open video-heavy apps because they lack a 'safety net' for their airtime.
- Mitigation Strategy: The platform will implement a 'Data Transparency Module' which provides a forecast of MB usage per lesson. Users will have the power to lock the app into 'Low-Data Mode' manually, ensuring the adaptive features never exceed their financial limits.

6.2 Reliability and Progress Continuity

A staggering 90% of users have abandoned a learning session due to poor internet. This creates a cycle of hesitation where users assume a platform will fail during a critical moment (like a quiz or an exam).

- Reluctance Factor: There is a deep skepticism regarding the stability of a new startup solution. Users worry that their time will be wasted if the app crashes or the connection drops mid-way through a task.
- Mitigation Strategy: Addressing the 92% user demand for auto-saving, we use local-first caching. The platform will feature a 'Live-Save' indicator that confirms progress is stored on the device every few seconds. This guarantees that even if the internet fails, the effort is preserved.

6.3 Technical Performance and Device Integrity

Since 83% of users learn via smartphones, the app’s impact on device health is a major consideration. Users are often wary of 'Smart' apps that might be heavy or poorly optimized.

- Reluctance Factor: Users often doubt the efficiency of a new platform, suspecting it may cause their phone to lag, drain battery rapidly, or occupy excessive storage space.
- Mitigation Strategy: We focus on a 'Lean-Build' approach (under 20MB). By processing adaptive transitions server-side rather than on the phone, we minimize CPU usage and battery heat, proving the app is built specifically for the mobile-first student.

6.4 Operational Skepticism & Control

While 97% of users want content that adjusts automatically, there is a natural skepticism regarding AI decision-making. Users do not want to feel 'trapped' by an algorithm.

- Reluctance Factor: Users hesitate when they feel they cannot override a system's decision (e.g., if the app switches to audio but the user still wants to see a low-quality diagram).
- Mitigation Strategy: Our 'Smart Interface' always provides a one-tap override. When the system adapts to bandwidth loss, it notifies the user and offers an immediate manual correction button. This balances automation with user authority.

6.5 Summary: Overcoming Reluctance through Trust

Area of Reluctance	Underlying Barrier	Strategic Solution
Economy	Expensive Data (44%)	Data Forecasting & Caps
Continuity	Abandonment (90%)	Local-First Auto-Save
Stability	Network Failures (97%)	3-Tier Adaptive Content
Accessibility	Offline Need (4.3/5)	Modular Downloads

7. Conclusion

The requirements gathering phase has made one thing very clear: in our current environment, a stable internet connection is the exception, not the rule. The data from our 59 respondents shows that while there is a strong desire to learn online, infrastructure failures and high data costs lead to a massive 90% abandonment rate.

By identifying these technical and economic barriers early in the **SDLC**, we have shifted from a general e-learning concept to a specific, network-aware design. Moving into the development phase, our priority is to build a mobile application that doesn't just work with the internet, but actively works *around* its limitations.

Addressing these frustrations now—through features like adaptive tiering and offline-first functionality—is the only way to ensure this platform becomes a reliable tool for students. We are now moving forward with a clear roadmap to create an inclusive learning experience that remains accessible regardless of a user's data budget or network strength.